Tempus

Introduction

- Why Tempus ?
- What we can do in Tempus ?

Timing Analysis

- Why is timing analysis important when designing a chip ?
- What are the types of timing analysis?
  - STA
  - DTA
- Why do we normally do STA rather than DTA
- What are the Timing violations we come across ?

Design Import

- Input Requirements
  - Timing Libraries (Liberty dotlib files)
  - Verilog Netlist
  - SDC Constraints
  - Parasitic Data (SPEF)
- Design Import Flow
  - Read all the inputs
  - Link design
  - Update timing
  - Generate reports
- Performing Sanity Checks
  - check_design
  - check_timing

Types of analysis

What is Mode?

What is Corner?

What is view?

Types are:

- Single mode analysis
- MMMC (Multi mode Multi Corner)
    - C-MMMC
    - D-MMMC

run.tcl

```
################################################################
# Purpose:  Show a simple Tempus STA script
# Purpose:  Hightlight the order of operations and simple commands
# Author:   John Schritz   Nov 2015
################################################################

################################
# Setup threading and client counts
################################
set_multi_cpu_usage -localCpu 8

################################
# Setup some global variables or report settings
################################
set_table_style -no_frame_fix_width -nosplit

################################
# Read the libraries
################################
read_lib     ../../libs/liberty/FreePDK45_lib_v1.0_worst.lib
read_lib     ../../libs/MACRO/LIBERTY/pllclk.lib
read_lib     ../../libs/MACRO/LIBERTY/ram_256x16A.lib   ../../libs/MACRO/LIBERTY/rom_512x16A.lib

################################
# Read the netlist in a gzipped format
################################
read_verilog "../../design/ECO_INIT_11_optSetup.enc.dat/dtmf_recvr_core.v.gz" ; # 8269 instances

################################
# Link the design
################################
set_top_module dtmf_recvr_core -ignore_undefined_cell

################################
# Check the size of the testcase
################################
```

```
#################################
set cellCnt [sizeof_collection [get_cells -hier *]]
puts "Your design has: $cellCnt instances"

#################################
# Load netlist parasitics
#################################
read_spef  ../../design/SPEF/corner_worst_CMAX.spef.gz

#################################
# Add constraints
#################################
read_sdc ../../design/dtmf_recvr_core.pr.sdc

#################################
# Adjust timer settings
#################################
set_delay_cal_mode -siAware true    ;# Turn on SI when true

#################################################################
#To dump aggressor info in report_delay_calculation -si command
##################################################################

set_si_mode -enable_delay_report true

###########################################
#enable the glitch reports to be generated
###########################################
set_si_mode -enable_glitch_report true

##########################
#Enable glitch propagation
##########################
set_si_mode -enable_glitch_propagation true

###################################
# Run timing
```

```
###################################
update_timing -full

###################################
# Run reports
###################################
report_timing

###################################
set reportDir newreports
file mkdir $reportDir
source ../scripts/reports.tcl

Puts "All done"
###################################
# If in interactive session, return to the Tempus prompt
# If in batch session, return to the Linux prompt
###################################
#exit
```

Reports.tcl

```
#########################
#Generate SI glitch report
#########################
report_noise -txtfile ${reportDir}/glitch.rpt

#####################
# Reports that check design health
#####################
#check_design -type timing -out_file    ${reportDir}/check_design.rpt
check_design -all > ${reportDir}/check_design.rpt
check_timing -verbose > ${reportDir}/check_timing.rpt
report_annotated_parasitics > ${reportDir}/annotated.rpt
report_analysis_coverage > ${reportDir}/coverage.rpt

#####################
# Reports that describe constraints
#####################
report_clocks                    > ${reportDir}/clocks.rpt
report_case_analysis             > ${reportDir}/case_analysis.rpt
report_inactive_arcs             > ${reportDir}/inactive_arcs.rpt

#####################
# Reports that describe timing health
#####################
report_constraint -all_violators                            > ${reportDir}/allviol.rpt
report_analysis_summary                                     > ${reportDir}/analysis_summary.rpt
report_timing -path_type summary_slack_only -late -max_paths 5  > ${reportDir}/start_end_slack.rpt

#####################
# GBA Reports that show detailed timing
#####################
report_timing -late   -max_paths 50 -nworst 1 -path_type full_clock -net  > ${reportDir}/worst_max_path.rpt
report_timing -early  -max_paths 50 -nworst 1 -path_type full_clock -net  > ${reportDir}/worst_min_path.rpt
report_timing -path_type end_slack_only                       > ${reportDir}/setup_1.rpt
report_timing -path_type end_slack_only  -early               > ${reportDir}/hold_1.rpt
report_timing -late    -max_paths 100                         > ${reportDir}/setup_100.rpt.gz
report_timing -early   -max_paths 100                         > ${reportDir}/hold_100.rpt.gz

#####################
# PBA Reports that show detailed timing
#####################
report_timing -retime path_slew_propagation -max_paths 50 -nworst 1 -path_type full_clock    > ${reportDir}/pba_50_paths.rpt
```

Command: report_constraints -all_violators -early

```
min_delay/hold
---------------------------
      ------------------------------------------------------------------------------------------------
       End Point                                  Slack      Cause
      ------------------------------------------------------------------------------------------------
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[10]/D r   -2.153     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[9]/D r   -2.140     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[11]/D r   -2.097     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[2]/D r   -2.085     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[13]/D r   -2.085     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[12]/D r   -2.072     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[15]/D r   -2.059     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[4]/D r   -2.041     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[6]/D r   -2.040     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[5]/D r   -2.036     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[7]/D r   -2.023     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[3]/D r   -2.018     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[0]/D r   -2.015     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[14]/D r   -2.002     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[8]/D r   -1.985     VIOLATED
      TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[1]/D r   -1.956     VIOLATED
      TDSP_CORE_INST/DATA_BUS_MACH_INST/data_out_reg[3]/SI r   -1.649     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[11]/D r   -1.451     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[2]/D r   -1.434     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[6]/D r   -1.431     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[8]/D r   -1.431     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[13]/D r   -1.427     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[10]/D r   -1.407     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[12]/D r   -1.405     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[9]/D r   -1.398     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[7]/D r   -1.394     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[14]/D r   -1.392     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[1]/D r   -1.381     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[3]/D r   -1.346     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[0]/D r   -1.320     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[15]/D r   -1.313     VIOLATED
      TDSP_CORE_INST/PORT_BUS_MACH_INST/data_out_reg[5]/D r   -1.201     VIOLATED
```

Command: report_constraints -all_violators -drv_violation_type max_transition

```
Check type : pulse_width
---------------------------
 No violating Checks with given description found

Check type : max_transition
---------------------------
     ----------------------------------------------------------------------------
      Pin Name                Required         Actual            Slack
     ----------------------------------------------------------------------------
     RAM_256x16_TEST_INST/RAM_256x16_INST/CLK   1.000              3.599             -2.599
     RAM_128x16_TEST_INST/RAM_128x16_INST/CLK   1.000              3.500             -2.500
     ROM_512x16_0_INST/A[2]   1.000              1.430             -0.430
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[10]   1.000              1.394             -0.394
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[7]   1.000              1.360             -0.360
     ROM_512x16_0_INST/A[1]   1.000              1.351             -0.351
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[4]   1.000              1.313             -0.313
     RAM_256x16_TEST_INST/RAM_256x16_INST/D[0]   1.000              1.229             -0.229
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[13]   1.000              1.179             -0.179
     RAM_256x16_TEST_INST/RAM_256x16_INST/D[3]   1.000              1.164             -0.164
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[12]   1.000              1.138             -0.138
     ROM_512x16_0_INST/CLK 1.000              1.115             -0.115
     ROM_512x16_0_INST/A[5]   1.000              1.104             -0.104
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[0]   1.000              1.084             -0.084
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[1]   1.000              1.027             -0.027
     RAM_256x16_TEST_INST/RAM_256x16_INST/D[12]   1.000              1.020             -0.020
     ROM_512x16_0_INST/A[6]   1.000              1.019             -0.019
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[8]   1.000              1.006             -0.006
     RAM_128x16_TEST_INST/RAM_128x16_INST/A[0]   1.000              1.002             -0.002

Check type : min_transition
---------------------------
No Violations found
```

Related Commands

- To display DRV violations
  - report_constraints -all_violators -drv_violation_type max_capacitance
  - report_constraints -all_violators -drv_violation_type min_transition
  - report_constraints -all_violators -drv_violation_type min_capacitance
  - report_constraints -all_violators -drv_violation_type max_fanout
  - report_constraints -all_violators -drv_violation_type min_fanout
- To display setup violations
  - report_timing -check_type setup
  - report_timing -late
  - report_constraints -all_violators -late
- To display hold violations
  - report_timing -check_type hold
  - report_timing -early
  - report_constraints -all_violators -early

Fixing Techniques

Setup : Take all violated paths and analyse the cell delays on data path.

- Upsize
- Vt swapping (HVT to LVT)
- Adding a buffer

Hold : Take all hold violated endpoints and add delay buffer with respect to negative slack if slack is more then we add more delay buffers .

- Downsize
- Vt swapping (LVT to HVT)

Tran : If there is more load on particular net, then we need better transition to drive the net. So we take the net driver and add one appropriate buffer on driver input side to improve transition.

Cap : If particular net having more fanout cells than the available load that impacts a cap violations in other words charging and discharging takes more time . So we can split the load to perform a cloning or add buffer at the driver output side.

Fixing of violations through ECO Opt

The only command to fix eco timing db is **write_eco_opt_db.** By using this design it automatically create ecoTimingDB.

Script to fix hold violations

- set_eco_opt_mode -verbose true
- set_eco_opt_mode -load_eco_opt_db ecoTimingDB
- eco_opt_design -hold

Script to fix DRC violations

- set_eco_opt_mode -verbose true
- set_eco_opt_mode -load_eco_opt_db ecoTimingDB
- set_eco_opt_mode -along_route_buffering true
- eco_opt_design -drv

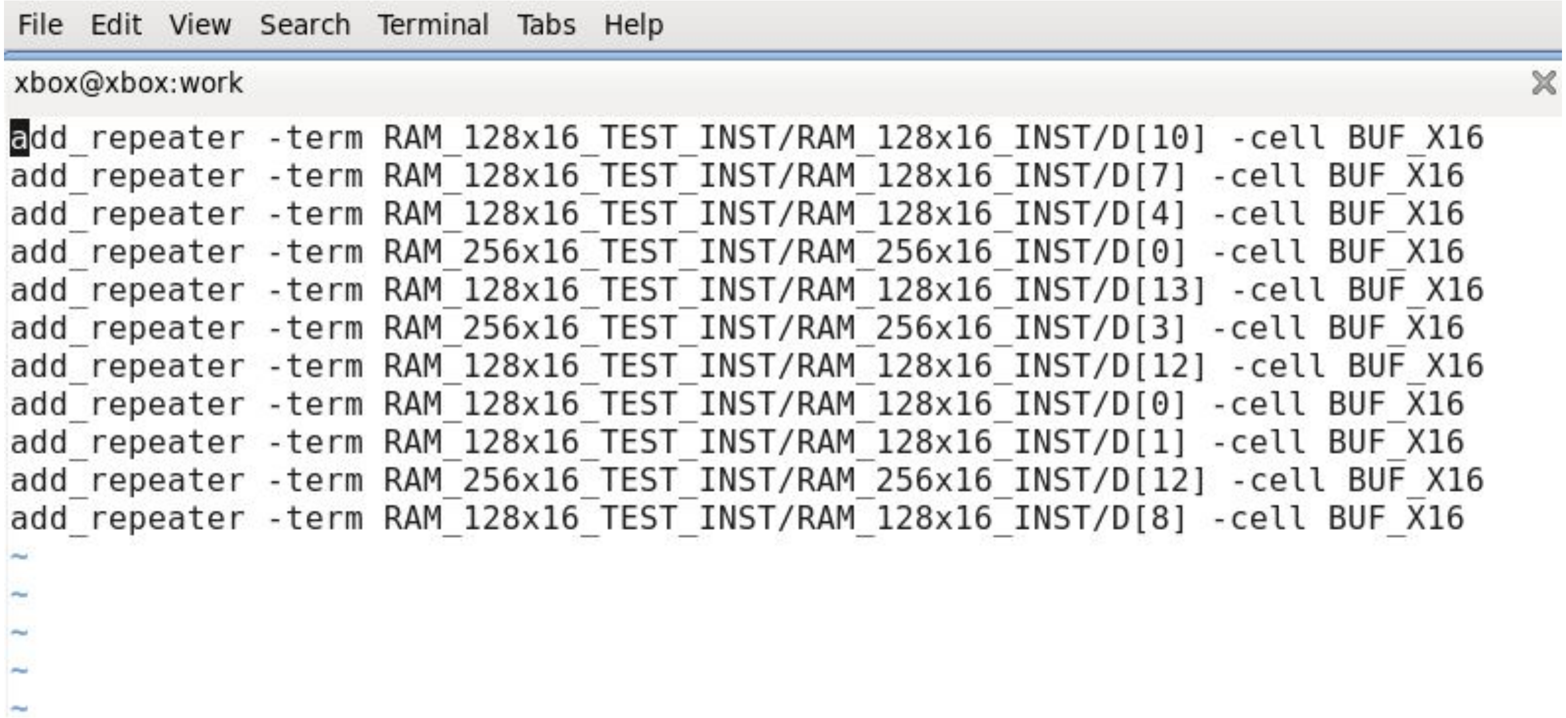
We can observe two files are created

- eco_innovus.tcl -> Given to PNR team
- eco_tempus.tcl

Fixing of DRVs manually

```
Check type : pulse_width
---------------------------
 No violating Checks with given description found

Check type : max_transition
---------------------------
     ----------------------------------------------------------------------------
      Pin Name                Required          Actual            Slack
     ----------------------------------------------------------------------------
     RAM_256x16_TEST_INST/RAM_256x16_INST/CLK    1.000              3.599              -2.599
     RAM_128x16_TEST_INST/RAM_128x16_INST/CLK    1.000              3.500              -2.500
     ROM_512x16_0_INST/A[2]    1.000             1.430              -0.430
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[10]    1.000              1.394              -0.394
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[7]    1.000              1.360              -0.360
     ROM_512x16_0_INST/A[1]    1.000             1.351              -0.351
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[4]    1.000              1.313              -0.313
     RAM_256x16_TEST_INST/RAM_256x16_INST/D[0]    1.000              1.229              -0.229
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[13]    1.000              1.179              -0.179
     RAM_256x16_TEST_INST/RAM_256x16_INST/D[3]    1.000              1.164              -0.164
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[12]    1.000              1.138              -0.138
     ROM_512x16_0_INST/CLK 1.000              1.115              -0.115
     ROM_512x16_0_INST/A[5]    1.000             1.104              -0.104
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[0]    1.000              1.084              -0.084
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[1]    1.000              1.027              -0.027
     RAM_256x16_TEST_INST/RAM_256x16_INST/D[12]    1.000              1.020              -0.020
     ROM_512x16_0_INST/A[6]    1.000             1.019              -0.019
     RAM_128x16_TEST_INST/RAM_128x16_INST/D[8]    1.000              1.006              -0.006
     RAM_128x16_TEST_INST/RAM_128x16_INST/A[0]    1.000              1.002              -0.002

Check type : min_transition
---------------------------
No Violations found
```

more balatran8_7_19 | grep "/D" | awk '{print “add_repeater -term ”$1 “ -cell BUF_X16”}'




```
add_repeater -term RAM_128x16_TEST_INST/RAM_128x16_INST/D[10] -cell BUF_X16
add_repeater -term RAM_128x16_TEST_INST/RAM_128x16_INST/D[7] -cell BUF_X16
add_repeater -term RAM_128x16_TEST_INST/RAM_128x16_INST/D[4] -cell BUF_X16
add_repeater -term RAM_256x16_TEST_INST/RAM_256x16_INST/D[0] -cell BUF_X16
add_repeater -term RAM_128x16_TEST_INST/RAM_128x16_INST/D[13] -cell BUF_X16
add_repeater -term RAM_256x16_TEST_INST/RAM_256x16_INST/D[3] -cell BUF_X16
add_repeater -term RAM_128x16_TEST_INST/RAM_128x16_INST/D[12] -cell BUF_X16
add_repeater -term RAM_128x16_TEST_INST/RAM_128x16_INST/D[0] -cell BUF_X16
add_repeater -term RAM_128x16_TEST_INST/RAM_128x16_INST/D[1] -cell BUF_X16
add_repeater -term RAM_256x16_TEST_INST/RAM_256x16_INST/D[12] -cell BUF_X16
add_repeater -term RAM_128x16_TEST_INST/RAM_128x16_INST/D[8] -cell BUF_X16
~
~
~
~
~
```

This is the file we need to source in our design for fixing of DRCs (max_tran) manually.

Multi-Mode Multi Corner

```
#Read MMMC file
################################
#
read_view_definition "/home/xbox/soundarya/pd/rak/Tempus181_STA_RAK/dmmmc/scripts/viewDefinition.tcl"
#
#################################
## Read the netlist in a gzipped format
#################################
#
read_verilog "../../design/dtmf_recvr_core.v.gz" ; # 8269 instances
#
#
#################################
## Link the design
#################################
set_top_module dtmf_recvr_core -ignore_undefined_cell
#
#################################
## Check the size of the testcase
#################################
set cellCnt [sizeof_collection [get_cells -hier *]]
puts "Your design has: $cellCnt instances"
#
#################################
## Load netlist parasitics
#################################
#
read_spef -rc_corner corner_worst_RCMAX "../../design/SPEF/corner_worst_RCMAX.spef.gz"
read_spef -rc_corner corner_worst_RCMIN "../../design/SPEF/corner_worst_RCMIN.spef.gz"

#
#################################################################
##To dump aggressor info in report_delay_calculation -si command
##################################################################
#
set_si_mode -enable_delay_report true
#
###########################################
```

```
###########################################
##enable the glitch reports to be generated
###########################################
set_si_mode -enable_glitch_report true
#
###########################
##Enable glitch propagation
###########################
set_si_mode -enable_glitch_propagation true
#
####################################
## Run timing
####################################
update_timing -full
#
Puts "All done"
####################################
## If in interactive session, return to the Tempus prompt
## If in batch session, return to the Linux prompt
####################################
##exit
```

```
##################################################
# Define the available libraries
##################################################
create_library_set -name libset_slow\
   -timing\
    [list ../../libs/liberty/FreePDK45_lib_v1.0_worst.lib\
    ../../libs/MACRO/LIBERTY/pllclk.lib\
    ../../libs/MACRO/LIBERTY/ram_256x16A.lib\
    ../../libs/MACRO/LIBERTY/rom_512x16A.lib]
create_library_set -name libset_fast\
   -timing\
    [list ../../libs/liberty/FreePDK45_lib_v1.0_typical.lib\
    ../../libs/MACRO/LIBERTY/pllclk.lib\
    ../../libs/MACRO/LIBERTY/ram_256x16A.lib\
    ../../libs/MACRO/LIBERTY/rom_512x16A.lib]

##################################################
# Define the available PVT corners
##################################################
create_rc_corner -name corner_worst_RCMAX\
   -preRoute_res 1.2\
   -postRoute_res {1.2 1.2 1.2}\
   -preRoute_cap 1.2\
   -postRoute_cap {1.2 1.2 1.2}\
   -postRoute_xcap {1.2 1.2 1.2}\
   -preRoute_clkres 0\
   -preRoute_clkcap 0
create_rc_corner -name corner_worst_RCMIN\
   -preRoute_res 0.8\
   -postRoute_res {0.8 0.8 0.8}\
   -preRoute_cap 0.8\
   -postRoute_cap {0.8 0.8 0.8}\
   -postRoute_xcap {0.8 0.8 0.8}\
   -preRoute_clkres 0\
   -preRoute_clkcap 0

##################################################
# Create the possible delay corners
```

```
####################################################
create_delay_corner -name delay_corner_slow_RCMAX\
   -rc_corner corner_worst_RCMAX\
   -early_library_set libset_slow\
   -late_library_set libset_slow
create_delay_corner -name delay_corner_fast_RCMAX\
   -rc_corner corner_worst_RCMAX\
   -early_library_set libset_fast\
   -late_library_set libset_fast
create_delay_corner -name delay_corner_slow_RCMIN\
   -rc_corner corner_worst_RCMIN\
   -early_library_set libset_slow\
   -late_library_set libset_slow
create_delay_corner -name delay_corner_fast_RCMIN\
   -rc_corner corner_worst_RCMIN\
   -early_library_set libset_fast\
   -late_library_set libset_fast

####################################################
# Define the available constraint types
####################################################
create_constraint_mode -name functional_mode -sdc_files [list /home/xbox/soundarya/pd/rak/Tempus181_STA_RAK/mmmc/work/changedpr.sdc]
create_constraint_mode -name scan_mode       -sdc_files [list /home/xbox/soundarya/pd/rak/Tempus181_STA_RAK/mmmc/work/changedscan.sdc]

####################################################
# Create the possbile views
# These are combinations of constraints, and delay corners
####################################################
create_analysis_view -name func_slow_RCMAX -constraint_mode functional_mode -delay_corner delay_corner_slow_RCMAX
create_analysis_view -name func_fast_RCMAX -constraint_mode functional_mode -delay_corner delay_corner_fast_RCMAX
create_analysis_view -name func_slow_RCMIN -constraint_mode functional_mode -delay_corner delay_corner_slow_RCMIN
create_analysis_view -name func_fast_RCMIN -constraint_mode functional_mode -delay_corner delay_corner_fast_RCMIN

create_analysis_view -name scan_slow_RCMAX -constraint_mode scan_mode -delay_corner delay_corner_slow_RCMAX
create_analysis_view -name scan_fast_RCMAX -constraint_mode scan_mode -delay_corner delay_corner_fast_RCMAX
create_analysis_view -name scan_slow_RCMIN -constraint_mode scan_mode -delay_corner delay_corner_slow_RCMIN
create_analysis_view -name scan_fast_RCMIN -constraint_mode scan_mode -delay_corner delay_corner_fast_RCMIN
```

```
##############################################
# Turn on the views that you want to use.
# Not all created views need to be used every
##############################################
set_analysis_view -setup [list \
func_slow_RCMAX \
func_fast_RCMAX \
func_slow_RCMIN \
func_fast_RCMIN \
scan_slow_RCMAX \
scan_fast_RCMAX \
scan_slow_RCMIN \
scan_fast_RCMIN \
] -hold [list \
func_slow_RCMAX \
func_fast_RCMAX \
func_slow_RCMIN \
func_fast_RCMIN \
scan_slow_RCMAX \
scan_fast_RCMAX \
scan_slow_RCMIN \
scan_fast_RCMIN \
]
```

- For fixing of Timing violations, need to set view analysis mode to one view.

  Commands:

  tempus 1 > all_analysis_views
  scan_fast_RCMIN func_slow_RCMAX scan_fast_RCMAX func_fast_RCMIN scan_slow_RCMIN func_fast_RCMAX scan_slow_RCMAX func_slow_RCMIN
  tempus 2> set_analysis_view -setup func_slow_RCMAX -hold func_slow_RCMIN
  tempus 3> all_setup_analysis_views
  func_slow_RCMAX
  tempus 4> all_hold_analysis_views
  func_slow_RCMIN
  tempus 5> report_constraint -all_violators -view func_slow_RCMIN -early
  tempus 6> report_timing -view func_slow_RCMIN -early

- The script below shows all violators based on all views

```
set hh [all_analysis_views ]
foreach  nn $hh {
report_constraint -view $nn -all_violators
}
```

- Here all the violators are fixed through ECO optimization techniques.

Initial Summary

Hold Analysis :

Views	WNS	TNS	VP	Worst End-Point
func_slow_RCMAX	-2.159	-58.348	35	TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[10]/D
func_fast_RCMAX	-1.300	-24.323	33	TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[10]/D
func_fast_RCMIN	-1.070	-17.029	27	TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[2]/D
func_slow_RCMIN	-1.885	-47.266	35	TDSP_CORE_INST/PROG_BUS_MACH_INST/data_out_reg[2]/D
scan_slow_RCMAX	-1.675	-25.134	19	TDSP_CORE_INST/DATA_BUS_MACH_INST/data_out_reg[3]/SI
scan_fast_RCMAX	-0.632	-4.727	17	TDSP_CORE_INST/DATA_BUS_MACH_INST/data_out_reg[3]/SI
scan_slow_RCMIN	-1.334	-17.783	19	TDSP_CORE_INST/DATA_BUS_MACH_INST/data_out_reg[3]/SI
scan_fast_RCMIN	-0.395	-0.603	11	TDSP_CORE_INST/DATA_BUS_MACH_INST/data_out_reg[3]/SI

Setup Analysis :

Views	WS	TNS	VP	Worst End-Point
func_slow_RCMAX	1.031	0.000	0	RESULTS_CONV_INST/r1477_reg[10]/D
func_fast_RCMAX	4.452	.0.000	0	RESULTS_CONV_INST/r1477_reg[10]/D
func_fast_RCMIN	5.942	0.000	0	RESULTS_CONV_INST/r852_reg[11]/D
func_slow_RCMIN	3.488	0.000	0	RESULTS_CONV_INST/r1477_reg[10]/D
scan_slow_RCMAX	1.031	0.000	0	RESULTS_CONV_INST/r1477_reg[10]/D
scan_fast_RCMAX	4.452	0.000	0	RESULTS_CONV_INST/r1477_reg[10]/D
scan_slow_RCMIN	3.488	0.000	0	RESULTS_CONV_INST/r1477_reg[10]/D
scan_fast_RCMIN	5.942	0.000	0	RESULTS_CONV_INST/r852_reg[11]/D

Initial Summary:

	Setup	Hold
Worst Slack	1.031	-2.159
Total Slack	0.000	-58.348
Violating paths	0	35

Final Summary:

	Setup	Hold
Worst Slack	1.031	0.000
Total Slack	0.000	0.000
Violating paths	0	0

```
set pp [open "/home/xbox/soundarya/pd/rak/Tempus181_STA_RAK/mmmc/work/ecotempus.tcl" w]
set hh [open "/home/xbox/soundarya/pd/rak/Tempus181_STA_RAK/mmmc/work/endpoint" r]
set end [read $hh]
foreach jj $end {
#puts $jj
set bb [get_property [report_timing -collection -to $jj -early ] slack]

if {$bb < 0 && $bb >= -1 } {
   puts $pp "add_repeater -cell BUF_X16 -term $jj"
   puts $pp "add_repeater -cell BUF_X8 -term $jj"

} elseif {$bb < -1 && $bb >= -2 } {
   puts "$pp add_repeater -cell BUF_X8 -term $jj"
 puts "$pp add_repeater -cell BUF_X8 -term $jj"

} else {
   puts $pp "add_repeater -cell BUF_X4 -term $jj"
   puts $pp "add_repeater -cell BUF_X4 -term $jj"
   puts $pp "add_repeater -cell BUF_X4 -term $jj"
   puts $pp "add_repeater -cell BUF_X8 -term $jj"
}
}
#set s [write_eco -output ecofile -format tempus]
#puts $s
close $pp
close $hh
```

This is the script for fixing hold violations automatically based on endpoints with respect to slack

SI Analysis

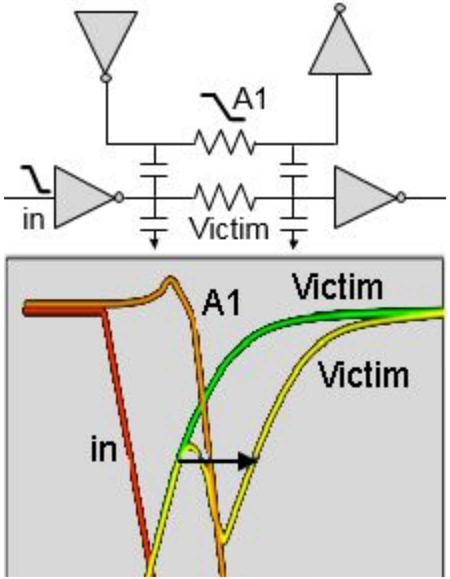
Tempus performs the following signal integrity analysis operations:

- SI Delay Analysis
- SI Glitch Analysis

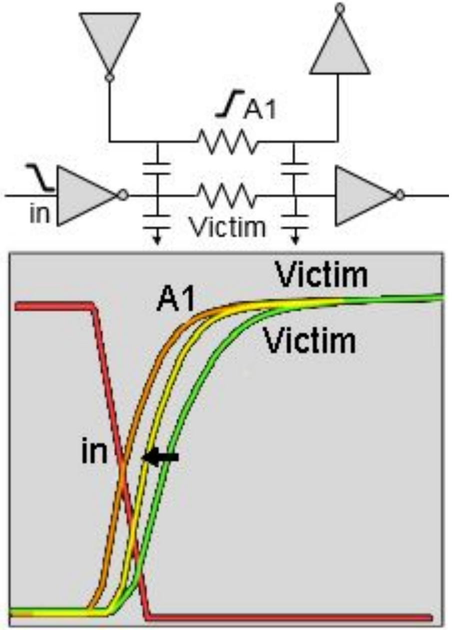
SI Delay Analysis :

- SI can increase or decrease signal delay, which can, in turn cause setup or hold failures.
- Consider that attacker A1 switches in the opposite direction to the victim, as shown in the figure below, then there can be a potential increase in the victim delay.
- SI can also decrease the delay and cause hold time failures. If both the attacker and victim are switching in the same direction, as shown in the figure below, then there is a decrease in the victim net delay. If this occurs on a critical minimum delay path, it can lead to hold violations.

SI Delay increases



SI Delay decreases



SI enabling switches

- set_si_mode –individual_attacker_threshold 0.015
  (Set the individual attacker threshold)

- set_si_mode –num_si_iteration 3 (Perform 3 SI iterations)

- set_si_mode -separate_delta_delay_on_data true -enable_delay_report true
  (Separate delta delay on data)

- set_delay_cal_mode –siAware true (Enable SI-aware delay calculation)

- report_timing   (for report the timing violations)

- report_delay_calculation –si –from inst1/A –to inst1/Y
  (Report details of SI delay calculation for an arc)

- report_noise –delay {min | max}   (report delay uncertainty due to SI)

SI enabling switches

- set_delay_cal_mode –siAware true
  (Enable signal integrity (SI) analysis)

- set_si_mode –enable_glitch_report true
  (Perform glitch analysis and enable the glitch reports to be generated)

- set_si_mode -enable_glitch_propagation true
  (enable glitch propagation)

- update_glitch (perform glitch analysis)

- report_noise -txtfile glitch.txt (generate SI glitch report)

Fixing of SI glitch through ECOs

- The below example shows fixing Si glitch violations only:

  set_eco_opt_mode –fix_max_cap false –fix_max_tran false –fix_glitch true

- The below example shows fixing of SI glitch violations using DRV fixing in addition to regular max_tran/max_cap violations:

  set_eco_opt_mode -fix_glitch true
  eco_opt_design -drv

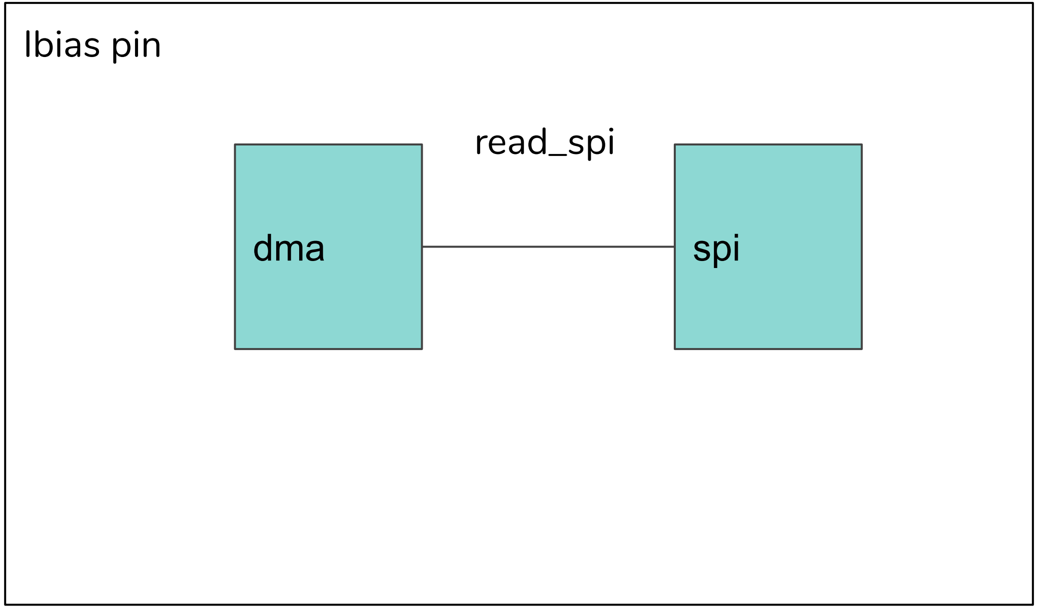
ECO generated glitch files

```
# eco_opt_design -drv
# set_eco_opt_mode settings :
#        -load_eco_opt_db "ecoTimingDB"
#        -fix_max_tran false
#        -fix_max_cap false
#        -fix_glitch true
# ====================

set_eco_mode -updatetiming false
report_resource
set_eco_mode -esomode true -prefixname ESO
change_cell -inst {TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p8465A} -cell AOI22_X4
add_repeater -cell BUF_X16 -name FE_ESOC0_n_269 -term {  TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p8465A/ZN } -relativeDistToSink 1
report_resource
set_eco_mode -esomode false

report_resource
update_timing -full
~
~
~
~
~
```

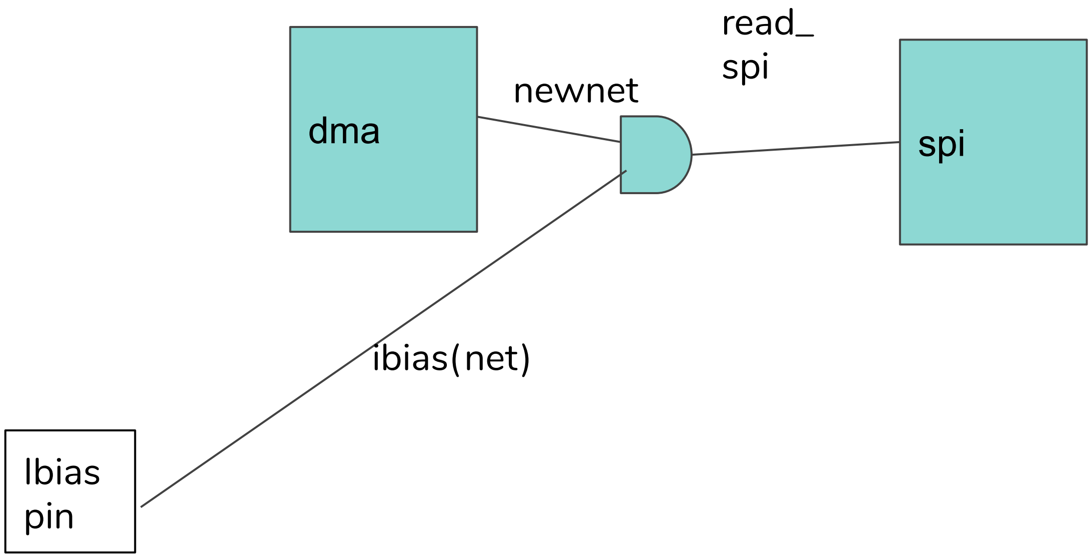
This is the file generated for innovus team (PNR)

```
# eco_opt_design -drv
# set_eco_opt_mode settings :
#        -load_eco_opt_db "ecoTimingDB"
#        -fix_max_tran false
#        -fix_max_cap false
#        -fix_glitch true
# =====================

report_resource
setEcoMode -updateTiming false
setEcoMode -refinePlace false -prefixName ESO -batchMode true -honorDontUse false -honorDontTouch false -honorFixe
catch { setEcoMode -honorFixedNetWire false }
ecoChangeCell -inst {TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p8465A} -cell AOI22_X4
ecoAddRepeater -cell BUF_X16 -name FE_ESOC0_n_269 -hinstGuide TDSP_CORE_INST/TDSP_CORE_GLUE_INST -term {  TDSP_COR
oSink 1
report_resource
setEcoMode -batchMode false

report_resource
refinePlace -preserveRouting true -hardFence false
~
~
~
```

Functional ECO



Adding a functional cell in between modules



Commands to add Functional ECO

- detach_net read_spi SPI_INST/p214748365A/B1
- add_inst ecoAND FreePDK45_lib_v1.0/AND2_X1
- add_net balanet
- attach_net balanet ecoAND/A1
- attach_net balanet SPI_INST/p214748365A/B1
- attach_net read_spi ecoAND/ZN
- attach_net ibias ecoAND/A2

By using above commands we can add a cell in between modules. This we called it as Functional ECO

Conclusion